# Akarsh Raj



AI Solutions Architect | Available immediately | No visa sponsorship required

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AI Solutions Engineer with **7+ years** building, launching and scaling production AI/ML systems across **Fortune 10** and **Fortune 500** companies. Hands-on experience architecting **Agentic AI, RAG, retrieval, APIs and cloud deployments**, while working directly with CXO and domain stakeholders to turn ambiguous business problems into production software. Delivered AI solutions generating **multi-million dollar** impact and 30–40% process improvements across **finance, healthcare, insurance and life sciences**. Kaggle Top 1% Data Scientist (Global).

## Education

**Friedrich-Alexander-Universität Erlangen-Nürnberg** Oct 2023 - Sep 2026

M.Sc. Artificial Intelligence Erlangen (Germany)

**Thesis** – Enhancing one step Text-to-image diffusion models with knowledge graphs & RL like reward based noise optimization (Grade - 1.3)

**University of Petroleum & Energy Studies** July 2013 - May 2017

B.Tech. Computer Science (Grade - 83.2%) Dehradun (India)

## Experience

**Siemens Healthineers** Erlangen (Germany)

Agentic AI Engineer (Werkstudent) July 2024 – September 2026

- Architected a **multi-agent RAG system** for financial decision support and reporting using **LangGraph/LangChain**, combining retrieval and LLM reasoning with deterministic Python tool execution for numerical calculations and **hallucination mitigation**.
- Designed traceable tool execution and outputs for transparent enterprise decision support, with **Langfuse-based LLM tracing, monitoring and evaluation**.
- Partnered with domain stakeholders to translate reporting workflows into agent capabilities, iterating from business requirements to a **containerized** and **deployed** application on SAP BTP

**Evalueserve** New Delhi (India)

Lead AI Solutions Engineer April 2022 – July 2023

- **Built the Data Science function** from 0 to 13 AI engineers/data scientists while remaining hands-on in solution architecture and client delivery. **Partnered directly with CXO leadership** on roadmap, effort/impact prioritization and **commercialization of internal Conversational AI tools**.
- Infinity AI**:** Led design and development of a **RAG application** designing retrieval, ranking strategies from scratch on top of GPT models for abstract Q&A and information extraction from organizational documents, improving research capabilities by 30%+ (TAT and quality). Implemented **PII redaction** and scalable **OCR/table-extraction** pipelines for scanned documents.
- Canvass AI**:** Led design and development of an NLP system for **Competitive Intelligence team** web-crawling and analyzing search results (SERP-API), NER, n-gram analysis, theme extraction, increasing researcher efficiency by 40%+.
- Partnered with internal teams on product development and go-to-market, translating internal AI tools into client-facing offerings.

**ZS Associates** New Delhi (India)

Advanced Data Science Associate Consultant October 2018 – March 2022

- Managed multiple analytics engagements and concurrent workstreams, **partnering with client stakeholders** to structure business problems, translate requirements into analytical solutions, and communicate recommendations.
- **Copay Anomaly Detection:** Built an anomaly detection engine (autoencoders, isolation forest) to identify **fraudulent pharmacies and prescribers**, uncovering **$3M+** in annual fraudulent transactions. Enhanced model interpretability with **SHAP explanations**. Deployed via Databricks and Azure.
- **Clinical Trial Optimization:** Translated an ambiguous, loosely-defined client ask into a structured problem statement. Designed a mixed-integer linear programming solution to **optimize clinical trial site selection and study duration**, satisfying thousands of operational constraints. Independently led a 4-hour client delivery call, presenting findings and defending recommendations directly to stakeholders.

- **Taxonomy Modeling:** Built an AI-driven **knowledge graph** from medical field agent notes using BERTopic, deploying a hierarchical taxonomy in Neo4j for semantic querying and relationship mapping.
- **Time Series Forecasting:** Forecasted **new-to-brand (NBRx)** drug volumes using Prophet, incorporating COVID-19 impact for improved accuracy. Also forecasted **enrolment rate** for a separate engagement.
- Mentored newly onboarded associates and managed delivery timelines and quality across multiple client engagements.

**Xceedance** New Delhi, India
Data Science Associate May 2017 – September 2018

- First year **loss prediction model** for one of the largest US based insurance firm which reaped them **4M+ profit**, annually. Supplemented finding with explainability models for stakeholder communication. Also developed an API using Flask for the same.
- Introduced **Text Field Classification System** assisting **Underwriting** process, minimizing process time **from 20 Minutes/Entry to an instant** using Python, trained LSTM model, and **OCR techniques (for scanned & Digital PDFs)** for one of the **Fortune 10 client**.

## Technical Skills

**AI Product & Solution Design:** Product roadmaps, use-case prioritization, impact/effort assessment, requirements gathering, problem structuring & scoping, stakeholder management, CXO communication, commercialization & GTM, Agile delivery

**GenAI & Agentic AI:** Agentic AI, multi-agent architectures, RAG, LLM APIs, LangGraph, LangChain, prompt engineering, Langfuse, AI evaluation, hallucination mitigation

**AI Governance & Data:** Responsible & explainable AI, PII redaction, embeddings, semantic & hybrid retrieval, Pinecone, Neo4j, SQL

**Machine Learning & Analytics:** Python, PyTorch, scikit-learn, Transformers, NLP, anomaly detection, forecasting, mixed-integer optimization, knowledge graphs

**Cloud, Software & MLOps:** Azure, AWS, GCP, SAP BTP / Cloud Foundry, Databricks, Docker, CI/CD, Git, FastAPI, Flask, API design

**Leadership:** Cross-functional collaboration, client presentations, team building & mentoring, technical-to-business communication

## Honors & Awards

Project Champion 2019 & 2020 (ZS Associates)
- Awarded for the best project across all offices at ZS

Dream Team Award (Evalueserve)
- Awarded for Canvass AI at Evalueserve

Competition Expert (Kaggle)
- Kaggle Top 1% (Global) - Competition Tier, Opening Speaker at Kaggle Days Meetup (Delhi)

## Languages

English (Fluent/Native), German (B1 ongoing), Hindi (Native)